Element C: Presentation/Justification of Solution Design Prioritized Criteria/ Constraints

Criteria (Ranked by Priority)

High Priority

1. Structural Load Capacity

- ➢ Must support at least 10 lb load with less than 1-0.5 inch movement at the plate mounting point over 15 minutes.

  Source: Zookeeper requirements (plate mass 7-8 lbs, safety factor requested)

2. Secure Pole Attachment

- ➢ Must securely attach to 12cm * 12 cm or 4.5 in by 4.5in square steel poles without slipping (cannot slip more than 1-0.5 inches).

  Source: Zoo enclosure measurements and stakeholder requirements

3. Position Stability

- ➢ Must maintain stable positioning during X-ray procedure without significant movement or rotation when some external force is added(giraffe head contact).

  Source: Dr. Louden Wright/veterinary staff at zoo meet

4. Fast Adjustment Time

- ➢ Must get setup within 5-15 minutes to lessen animal stress/total procedure time.

  Source: set up requirements/email questions (15-20 min total for the X-ray to get set up)

Medium Priority

5. Vertical Adjustment Range

- ➢ Must allow vertical adjustment range of at least 2-3 feet to accommodate giraffe head and additionally leg imaging.

  Source: Enclosure geometry + veterinary imaging needs

6. Multi-Axis Rotation

- ➢ Should allow adjustment in at least 2-3 rotational axes with approximately ±90° range to properly align the X-ray plate.

  Source: Stakeholder feedback (alignment flexibility needed)

Low Priority

7. Non-Toxic, Neutral Materials/Finish

- ➢ Should use non-toxic materials and neutral colors to reduce animal stress and prevent harm from contact or licking.

  Source: Zoo safety practices/stakeholder recommendation

Constraints (Ranked by Priority)

High Priority

1. Class Budget

- ➢ Total project cost must not exceed $250.

  Source: E4USA/engineering class funding limit (Mr. Kiser/Mr. Keidel)

2. Project Timeline

- ➢ Must be completed by May 11th (final delivery) and design showcased by April 28th.

  Source: Course schedule

3. Material Restrictions

- ➢ Must use animal safe materials: wood, plastic, metal; no sharp/hazardous objects allowed

  Source: Zoo safety requirements/instructor guidelines

4. Animal Safety

- ➢ Device can't have sharp edges, or detachable parts injuring animals/zoo staff.

  Source: Veterinary safety standards

Medium Priority

5. Tool/Manufacturing Availability

- ➢ Design must be made using available school tools (drill, saw, hand tools, etc.)

  Source: E4USA lab constraints